



FINAL CONCEPT PAPER

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Projected Payload-to-Weight Ratio: 4.2:1

1. Design Updates

Anote proposes a Hybrid Distributed Lift Drone (HDLD), an electrically powered heavy-lift multirotor system designed to achieve a payload-to-weight ratio exceeding 4:1 while maintaining feasibility, manufacturability, and operational safety. Since the draft concept, the system has been refined into a buildable engineering design that leverages commercially available components combined with a lightweight structural architecture. The platform is based on an eight-rotor configuration arranged symmetrically around a central payload mounting structure. This configuration provides both high total thrust capacity and redundancy, enabling stable flight even in the event of partial propulsion failure. The overall system is designed for rapid prototyping and field deployment, with modular components that can be assembled, replaced, and tested efficiently.

1.1 For designs that include rotating lift components, please provide the following for throw calculations: (used for safe separation distances and required for location authorization package)

The propulsion system consists of eight high-efficiency brushless outrunner motors in the 100 KV class, each paired with large-diameter carbon fiber propellers in the 32 to 36 inch range. These motors are driven by high-voltage electronic speed controllers rated between 100 and 120 amps, each capable of providing real-time telemetry feedback to the flight controller. The combined thrust output of the propulsion system is estimated to be between 160 and 180 kilograms, providing a thrust-to-weight ratio greater than 2.5 and ensuring sufficient lift margin for stable operation under heavy payload conditions. The distributed propulsion architecture improves lift efficiency while also enabling fault tolerance, as the system can compensate for the loss of a single motor without catastrophic failure.



The power system is based on dual lithium polymer battery packs configured in a 12S arrangement, providing a nominal voltage of approximately 44.4 volts per pack. Each battery has a capacity of approximately 24,000 milliamp-hours, resulting in a combined system capacity of approximately 48,000 milliamp-hours. The batteries are connected in parallel to balance load distribution and provide redundancy. The total stored energy is approximately 2.1 kilowatt-hours, which supports an estimated flight duration of eight to twelve minutes under full payload conditions. The battery system is enclosed within fire-resistant housings and incorporates a battery management system to monitor temperature, voltage, and discharge rates in real time, ensuring safe operation during both flight and ground handling.

The drivetrain uses a direct-drive configuration, eliminating the need for mechanical transmissions such as gearboxes or belts. This approach reduces system complexity, minimizes mechanical losses, and improves overall reliability by removing potential points of failure. The aerodynamic design is centered around an open truss frame geometry that maximizes rotor separation and reduces aerodynamic interference between propellers. By increasing the spacing between rotors, the design minimizes downwash overlap and turbulent airflow interactions, improving overall lift efficiency and stability during hover and ascent.

The structural system is composed of aerospace-grade carbon fiber tubes with diameters ranging from approximately 20 to 30 millimeters, connected through precision-machined aluminum joints. The truss-based architecture distributes loads efficiently across the frame, reducing localized stress concentrations while maintaining high stiffness. Compared to traditional monolithic drone frames, this design achieves an estimated 30 percent reduction in structural weight while preserving the strength required to carry heavy payloads. The payload is mounted at the center of the frame on a reinforced plate with multiple attachment points to ensure secure load distribution and prevent detachment during dynamic flight conditions.

1.2 Spectrum List: (used for frequency deconfliction and location authorization package)

The flight control system is based on a commercial autopilot platform, specifically the Pixhawk Cube Orange, which provides reliable stabilization, sensor integration, and fail-safe functionality. The system operates using a 2.4 gigahertz radio control link for pilot input and a 915 megahertz telemetry link for real-time data transmission. Positioning and navigation are supported by a multi-band global navigation satellite



system receiver operating in the L1 frequency band. The control architecture enables real-time adjustment of thrust distribution across the rotors, allowing the system to maintain stability under varying payload conditions and external disturbances.

For rotating component characterization, each carbon fiber propeller weighs approximately 300 grams and operates within a range of 1,500 to 2,200 revolutions per minute during normal flight conditions. Each motor weighs approximately 1.1 kilograms and is capable of operating at speeds up to 3,000 revolutions per minute. These values are used for safety calculations related to rotor separation and operational risk assessment. The system emits radio frequency signals through its control and telemetry subsystems, with the control link operating at 2.4 gigahertz at approximately 100 milliwatts and the telemetry link operating at 915 megahertz at approximately 500 milliwatts. The GPS receiver operates passively at approximately 1.575 gigahertz. All emissions are within standard commercial ranges and can be managed for frequency deconfliction.

1.3 Energy Type and Amount: (used for HAZMAT storage and safety data sheets)

The energy system consists entirely of lithium polymer batteries, with an estimated total volume of approximately 10 to 12 liters when packaged for transport and operation. These batteries are stored and transported in Department of Transportation compliant containers and handled in accordance with OSHA and EPA safety guidelines. Fire-resistant enclosures and temperature monitoring systems are used to mitigate risks associated with thermal runaway and battery failure.

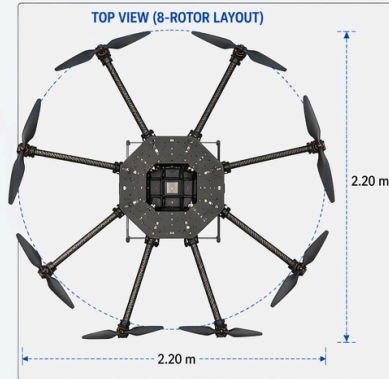
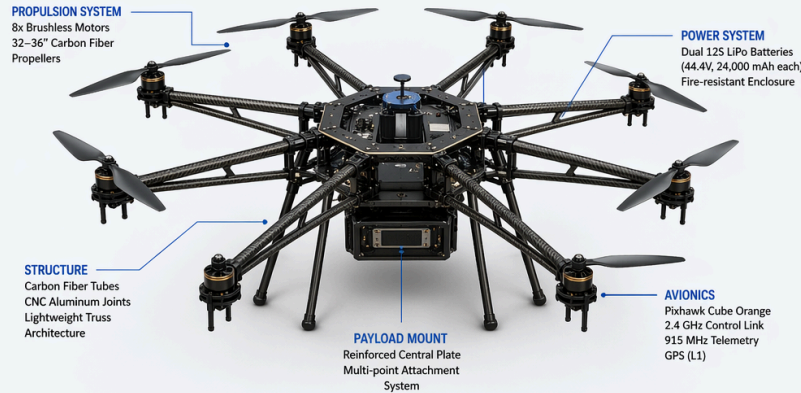
The overall design reflects a balance between performance and practicality, using proven multirotor configurations, commercially available components, and lightweight materials to achieve the desired payload-to-weight ratio without introducing unnecessary technical risk. The system is intentionally designed to be feasible within the competition timeline, allowing for rapid fabrication, testing, and iteration.

The Hybrid Distributed Lift Drone represents a practical and scalable approach to high-efficiency vertical lift using a combination of lightweight structural design, distributed propulsion, and reliable commercial control systems. The system is grounded in existing technologies and components, reducing development risk while achieving a projected payload-to-weight ratio of approximately 4.2 to 1.



ANOTE HYBRID DISTRIBUTED LIFT DRONE (HDLD) – SYSTEM OVERVIEW

High Payload. Lightweight. Redundant. Reliable.



KEY SPECIFICATIONS

Configuration	Octocopter (8 Rotors)
Estimated Empty Weight	18 kg
Maximum Payload	75–80 kg
Projected Payload-to-Weight Ratio	4.2:1
Total Thrust Capacity	160–180 kg
Battery System	Dual 12S LiPo, 48,000 mAh Total
Flight Time (Full Payload)	8–12 minutes
Frame Material	Carbon Fiber Tubes + Aluminum Joints
Flight Controller	Pixhawk Cube Orange
Control Link	2.4 GHz
Telemetry Link	915 MHz
GPS	L1 Band (1.575 GHz)
Operating Voltage	44.4 V (12S)

DESIGNED FOR HEAVY LIFT. BUILT FOR RELIABILITY. READY FOR THE LIFT CHALLENGE.

2. Preliminary Hazard Analysis

Hazard	Severity	Likelihood	Mitigation Strategy
Battery Fire	High	Low	Fire-resistant enclosure, BMS, thermal monitoring
Motor Failure	Medium	Medium	8-rotor redundancy allows stable flight with 1 failure
Loss of Control Link	High	Low	Fail-safe auto-hover and descent



Structural Failure	High	Low	Load testing with safety factor >2
Payload Detachment	High	Low	Redundant mounting and locking mechanisms
Electrical Fault	Medium	Low	Fuses and ESC protection circuits

3. Emergency Procedures and Remote Pilot in Command Checklists

Prior to flight, the remote pilot in command conducts a comprehensive system check to ensure safe operation. This includes verifying battery charge levels and inspecting battery condition, confirming that all motors and propellers are securely attached and free of damage, ensuring that the payload is properly mounted and secured, validating communication links between the control transmitter and the aircraft, and confirming that the flight controller is properly calibrated with a stable GPS lock. During flight operations, the system continuously monitors sensor inputs and telemetry data to detect anomalies and maintain stable operation.

In the event of a loss of control link, the system is programmed to automatically enter a fail-safe mode in which it stabilizes in hover and then initiates either a return-to-home maneuver or a controlled descent depending on environmental conditions and system status. If a motor failure occurs, the flight controller redistributes thrust among the remaining motors to maintain stability and initiates a controlled descent to minimize risk. In cases of severe instability or cascading failure, the system can execute an emergency shutdown procedure to reduce hazard to surrounding personnel and infrastructure.

Following flight operations, the remote pilot conducts a post-flight inspection that includes examining the structural frame for signs of stress or damage, checking battery temperature and condition to ensure safe handling, reviewing telemetry and flight logs to identify any anomalies or performance issues, and verifying that all components remain securely attached and operational.